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Roll No

EE-7001-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

High Voltage Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carries equal marks.

1. a) Explain high voltage technology. 7
b) Discuss need for generating high voltages in laboratory. 7
2. a) What is “thermal breakdown” in solid dielectrics, and how is it practically more significant than other mechanisms? 7
b) Describe how vacuum breakdown is different from normal breakdown of a gas. 7
3. a) What do you understand by “intrinsic-strength” of a solid dielectric? How does breakdown occur due to electrons in a solid dielectric? 7
b) What is a trigation gap? Explain its functions and operation. 7
4. a) What is a Tesla coil? How are damped high frequency oscillations obtained from a Tesla coil? 7
b) Why is a Cockcroft-Walton circuit performed for voltage multiplier circuits? Explain its working with schematic diagram. 7

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5. a) Explain the principle and construction of an electrostatic voltmeter for very high voltages. What are its merits and demerits for high voltage a.c. measurements? 7
b) Discuss the different methods of measuring high d.c voltages. What are the limitations in each method? 7
6. a) Explain how a sphere gap can be used to measure the peak value of voltages. What are the parameters and factors that influence such voltage measurement? 7
b) What is capacitance voltage transformer? Explain with phasor diagram how a tuned capacitance voltage transformer can be used for voltage measurements in power systems. 7
7. a) Explain the partial discharge tests on high voltage cables. How is a fault in the insulation located in this test? 7
b) What is the significance of impulse tests? Briefly explain the impulse testing of insulators. 7
8. Write short notes on any two of the following. 14
 - a) Bubble's theory
 - b) Streamer theory
 - c) Surge current measurement
